

Spatiotemporal Flood Risk Modelling Under a Single-Orbit SAR Constraint, Part I: Confound-Controlled Susceptibility and a Three-Stage Validation Ladder

Abstract—Over Dakshina Kannada, Karnataka, exactly one Sentinel-1 orbit covers the district on a strict twelve-day cycle, so no individual flood there can be mapped from synthetic aperture radar (SAR): a flash flood rises and recedes inside the revisit gap. This paper, the first of two, treats that constraint as the design premise and builds the static half of a two-stage risk model, $R(x, t) = S(x) \cdot T(t)$. A multi-temporal SAR flood-frequency inventory replaces the single-event map that the revisit interval forbids. A remote-sensing confound is then identified and removed: a model given land cover and NDVI reaches ROC-AUC 0.9878, but SHAP shows it has learned that closed canopy hides water from SAR, not flood physics. The terrain-and-rainfall model scores 0.9625 on a random split, 0.9474 under spatial-block cross-validation, and 0.871 transferring to a held-out city window never seen in training, where its highest-susceptibility tenth contains 49.5% of all SAR-observed flooding. We show the calibrated score is tied to the balanced training prior and must not be read as a flood probability without correction. The dynamic trigger $T(t)$, keyed to India Meteorological Department rainfall categories and fitted to no event, ranks two documented floods above ordinary and dry conditions but separates a flood from a heavy-rain non-flood day by only 0.011, a margin governed almost entirely by its antecedent-saturation term. Each stricter validation protocol lowers the headline number, and we report the descent rather than the first figure. Part II applies the resulting risk surface to emergency routing and dispatch.

Index Terms—Flood susceptibility, synthetic aperture radar, Sentinel-1, spatial cross-validation, explainable AI, probability calibration, Google Earth Engine.

I. INTRODUCTION

Flooding is the most frequently recurring natural hazard in India, and the country’s south-western coastal belt concentrates several of the mechanisms that make it difficult to forecast at a useful spatial scale. Dakshina Kannada district, centred on the port city of Mangaluru in coastal Karnataka, receives more than 4000 mm of rainfall in an average monsoon, drains two substantial rivers, the Nethravathi and the Gurupura, into a shared estuary, and does so across a coastal plain narrow enough that fluvial discharge and tidal stage interact within a few kilometres of the built-up area. Extreme-rainfall frequency over this stretch of the Western Ghats and the Karnataka coast has been shown to be trending upward [1], while land-use change in the Netravathi basin has altered its runoff response [2, 3] and the district’s land cover has itself changed measurably over recent decades [4], so the hazard is neither stationary nor purely meteorological. Two recent events frame this study: the August 2024 Nethravathi river flood and the

May 2025 Mangaluru urban flood. Existing flood work on this basin is predominantly static and index-based, for example the GIS and analytic-hierarchy susceptibility zoning of Nirmala [5], which produces a fixed map with no temporal component.

The dominant response in the literature is flood-susceptibility mapping: train a classifier on terrain and hydrological conditioning factors against an inventory of observed flooding, and produce a wall-to-wall map of relative propensity [6–9]. The approach is mature, and the recent generation pairs it with explainable-AI attribution so that the resulting map can be interrogated, not merely displayed [10–12]. Our work sits in this tradition, and departs from it in three specific ways that this paper’s contributions address.

a) The observational premise is usually left implicit:

Susceptibility studies typically source their inventory from whatever flood observations are available, then treat the resulting map as if it were a spatial model of flooding in general. But the density of the underlying satellite record constrains what such a map can honestly claim. Over Dakshina Kannada, Sentinel-1 [13] coverage consists of a *single* orbit geometry, a descending pass on relative orbit 63, giving a nominal revisit of approximately twelve days. This matters because rapid SAR inundation mapping of the kind demonstrated elsewhere in India [14] presumes an acquisition close enough in time to the peak to be representative. We verified this directly against the Copernicus archive instead of assuming the constellation’s global six-day figure applies. The consequence is strict: no individual flood in this district can be mapped from SAR, because the hydrograph of a coastal flash flood peaks and recedes well inside the revisit gap. Work that frames flood mapping as per-event spatial classification implicitly assumes a denser record than exists here. Section IV-A sets out the two-stage decomposition that follows from taking this seriously.

b) Validation is usually optimistic: Random k -fold cross-validation on geospatial samples leaks information across fold boundaries, because neighbouring points share terrain; the resulting score measures interpolation within sampled neighbourhoods, not skill at an unvisited location [15–17]. Reported AUCs above 0.95 are common in this literature and are rarely accompanied by a spatial control. We report random k -fold, spatial-block k -fold, and a strict geographic-transfer test in which the model is trained entirely outside the evaluation window (Section V-C).

c) Free parameters are usually asserted: Flood-aware routing formulations introduce cost-function constants, a

penalty weight and an impassability threshold, that are typically fixed by judgement and never swept. Because these constants determine which roads the router will refuse to use, they are exactly the parameters an operational deployment would be questioned on. Part II [18] reports the full 7×7 sweep and finds a result that is more nuanced than a stability claim.

A. Contributions of Part I

This paper is the first of a two-part study. Part I establishes the risk model and the evidence for trusting it; Part II [18] applies that model to emergency routing and dispatch over a real road network. The contributions of this part are:

- 1) A SAR flood-frequency inventory that is explicit about the single-orbit revisit limit and about what it therefore cannot claim.
- 2) Identification and removal of a remote-sensing confound in which land cover and NDVI let the model learn SAR detectability instead of flood hazard, quantified by ablation.
- 3) A three-stage validation ladder, random split to spatial-block cross-validation to geographic transfer, reporting the full descent of the headline metric rather than its first value.
- 4) A prior-correction analysis showing that a balanced-sample calibrated score is not a deployment-prevalence flood probability.
- 5) A temporal trigger built only from published rainfall categories, with its discriminative failure mode sought adversarially and reported.

Figure 2 maps the full two-part pipeline; the shaded stages are the subject of this paper.

II. RELATED WORK

A. Machine learning for flood susceptibility

Tree ensembles dominate the field. Tehrany et al. [6] established the SVM-and-frequency-ratio baseline; Khosravi et al. [7] benchmarked multi-criteria against machine-learning approaches; Arabameri et al. [8] and Saha et al. [9] established random forest and boosted-tree variants as the practical default. Reported AUCs cluster between 0.85 and 0.95. Wang et al. [19] and Avand et al. [20] extended the approach to data-scarce basins. The conditioning factors are largely standardised: elevation, slope, curvature, the topographic wetness index [21], height above nearest drainage [22, 23], distance to channel, drainage density and rainfall climatology.

Explainability entered the field recently. Lundberg and Lee [10] introduced SHAP; Aydin and Iban [11] and Seydi et al. [12] applied it to flood susceptibility, generally to confirm that elevation and channel proximity dominate. Our SHAP analysis is used differently, as a diagnostic that exposed a confound (Section IV-E), not as a post-hoc confirmation.

B. Spatial validation

That geospatial models are routinely over-validated is established but under-applied. Roberts et al. [15] formalised

block cross-validation for spatially structured data; Ploton et al. [16] demonstrated that ignoring it inflates reported accuracy in remote-sensing regressions; Meyer and Pebesma [17] extended the argument to area-of-applicability estimation. Flood-susceptibility papers reporting spatial CV remain a minority, which is part of the motivation for Section V-C.

C. SAR flood mapping

SAR change detection for inundation is well established [24–26], typically with backscatter thresholding against a dry reference and a permanent-water mask [27]. The recognised limitation is that C-band cannot detect water beneath closed canopy [28, 29], which is precisely the mechanism behind the confound we isolate. Revisit-limited inventories are usually handled by aggregating across many passes into a frequency product [27, 30], the approach adopted here. Closest to the present work, Sadhwani and Eldho [31] combine SAR with elevation data in a dual machine-learning scheme for inundation mapping *and* depth estimation over the Indian west-coast belt, using the same sensor and terrain inputs as this study. Their depth-regression stage is precisely the extension our Limitation 1 identifies as the highest-value next step, and the contrast is instructive: they target an absolute physical quantity, while the index reported here is explicitly relative.

D. Gap

No prior work we are aware of combines an inventory that is explicit about single-orbit revisit limits, a trigger validated against documented events without fitting, a routing layer with a complete parameter sweep and a search-transparent algorithm comparison, and a dispatch layer verified against the exact classical optimum, applied to one Indian coastal district with every number reproducible from public data.

III. STUDY AREA AND DATA

A. Study area

Dakshina Kannada district occupies the coastal plain and lower Western Ghats scarp of south-western Karnataka (Figure 1). The modelling domain is the bounding box $[74.65^\circ, 12.65^\circ]$ to $[75.40^\circ, 13.15^\circ]$; detailed routing and validation are conducted over the Mangaluru city window $[74.78^\circ, 12.80^\circ]$ to $[74.92^\circ, 13.00^\circ]$. Table I summarises the physical setting as measured from the assembled data stack, not quoted from secondary sources.

Three features of the setting drive the methodology. First, relief is extreme over short distances: elevation in the sampled floodable domain spans -5 m to 279 m, with the Ghats escarpment rising steeply inland, so runoff concentration times are short and flood peaks are flashy. Second, the coastal strip is low and tidally influenced, so the estuary can impound river discharge, a coupling that a rainfall-only trigger captures only indirectly. Third, land cover in the low-lying non-flooded fraction of the district is predominantly closed-canopy vegetation, which is what makes the SAR detectability confound of Section IV-E bite.

TABLE I: Study-area characteristics. Terrain and rainfall statistics are computed over the sampled floodable-lowland domain ($n = 3000$ points); network statistics are from the OpenStreetMap drivable graph of the Mangaluru window.

Property	Value	Source
District	Dakshina Kannada, Karnataka, India	FAO GAUL
Modelling bounding box	74.65°–75.40°E, 12.65°–13.15°N	–
Routing window	74.78°–74.92°E, 12.80°–13.00°N	–
Principal rivers	Nethravathi, Gurupura (shared estuary)	MERIT Hydro
Elevation range (sampled)	–5 m to 279 m	SRTM
HAND range (sampled)	0 m to 79.4 m	MERIT Hydro
Mean annual rainfall	3373 mm to 4401 mm (mean 4124)	CHIRPS 2020–2024
Monsoon regime	South-west, June–September	IMD
Sentinel-1 coverage	1 orbit (descending, rel. orbit 63)	Copernicus
Nominal SAR revisit	12 d	Copernicus
Road graph	11,910 nodes, 28,529 edges	OpenStreetMap
Flood-prone edges ($S > 0.5$)	4873 (17.1%)	this study
Hospitals modelled	7	OpenStreetMap
Incident sites modelled	7	this study

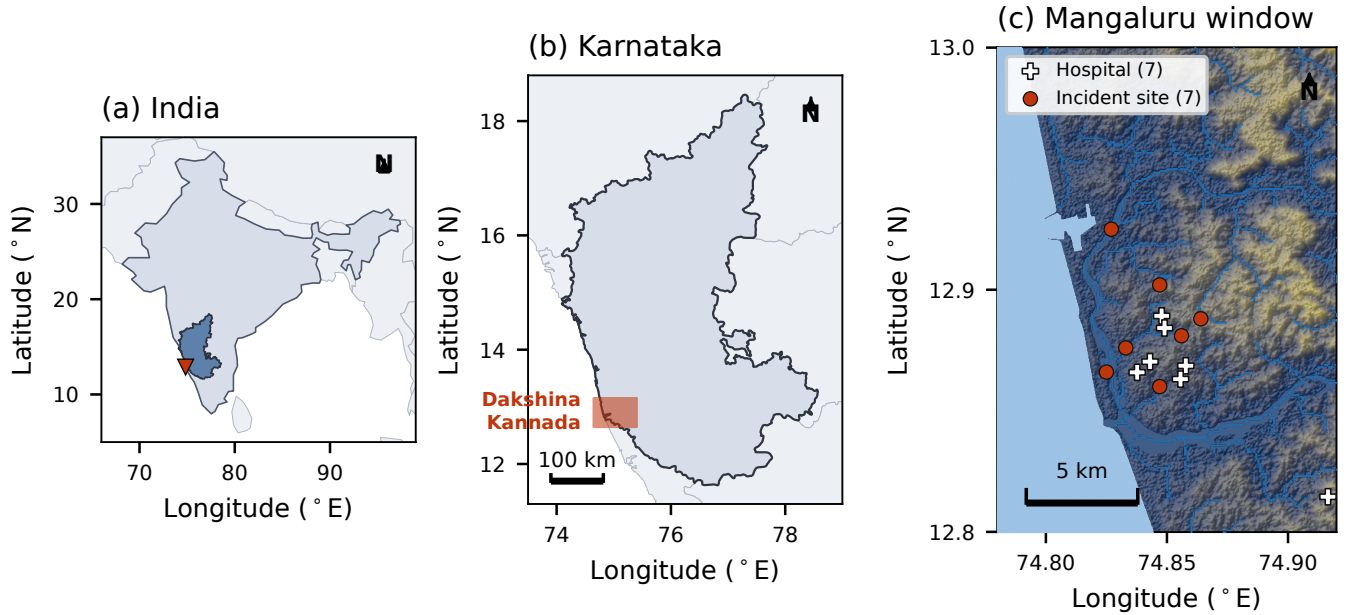


Fig. 1: Study area. (a) India with Karnataka shaded and Mangaluru marked. (b) Karnataka with the Dakshina Kannada modelling box. (c) The Mangaluru routing window: hillshaded SRTM terrain, the MERIT-derived channel network, the sea and estuary, and the seven hospitals and seven incident sites used throughout. The coastline is the valid-data boundary of the SRTM stack, not a cartographic overlay.

B. Documented events

Two events anchor the temporal validation. The August 2024 Nethravathi flood followed sustained catchment wetting, 216.9 mm over the preceding 72 h, with a comparatively modest 53.4 mm in the final 24 h. The May 2025 Mangaluru urban flood combined a high 24-hour total (87.2 mm, IMD “Heavy”) with a similarly saturated antecedent state (205.6 mm/72 h). Their contrasting structure, one saturation-driven, one intensity-driven, is what makes them a useful pair for testing a trigger that weights both. Full inputs are in Table IX.

C. Data sources

All inputs are public and ingested programmatically (Table II). Satellite and terrain layers are processed server-side on Google Earth Engine [32]; only sampled and aggregated outputs are downloaded.

IV. METHODOLOGY

A. Why the risk model is decomposed

A fully spatiotemporal flood model would estimate inundation $I(\mathbf{x}, t)$ jointly in space and time. Fitting such a model requires observations resolved at the timescale of the process.

TABLE II: Data sources. Every layer is public; none is hand-authored or simulated.

Source	Product	Resolution	Role
Sentinel-1 [13]	GRD IW, VV+VH, rel. orbit 63	10 m	Multi-temporal flood-frequency inventory
JRC GSW [27]	Water occurrence	30 m	Permanent-water exclusion mask
SRTM [33]	Digital elevation model	30 m	Elevation, slope, aspect, curvature
MERIT Hydro [34]	HAND, upstream area	90 m	HAND, TWI, distance to river, drainage density
CHIRPS [35]	Daily precipitation	5 km	Mean annual rainfall climatology
ESA WorldCover [36]	Land cover	10 m	Ablation only (excluded from primary model)
Sentinel-2	Surface reflectance	10 m	NDVI; ablation only
Open-Meteo / ERA5 [37]	Hourly precipitation	point	Live and archived antecedent rainfall
WorldTides	Tide height	point	Acquired for the live dashboard; <i>not</i> used in $T(t)$ (§V-F)
OpenStreetMap	Drivable road network	vector	Routing graph [38]
Natural Earth	Administrative outlines	10 m scale	Cartography only

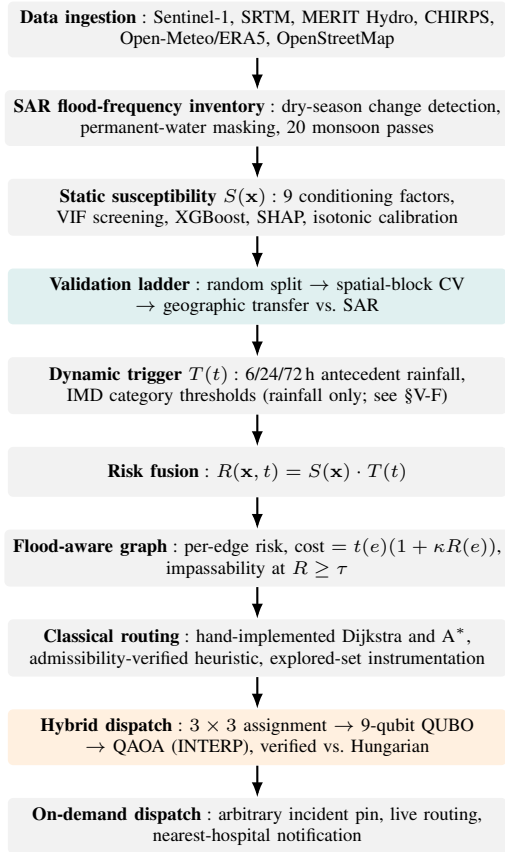


Fig. 2: Pipeline. The two-stage split into a static SAR-trained surface and a dynamic live trigger (shaded path through $S(\mathbf{x})$ and $T(t)$) is a direct consequence of the twelve-day single-orbit revisit established in Section I, not a modelling preference.

With one usable SAR orbit at a twelve-day cadence, the observational record cannot constrain the temporal dimension

at all. We therefore factor the problem:

$$R(\mathbf{x}, t) = S(\mathbf{x}) \cdot T(t), \quad S(\mathbf{x}) \in [0, 1], \quad T(t) \in [0, 1], \quad (1)$$

where $S(\mathbf{x})$ is a time-invariant relative propensity estimated from the *frequency* of observed flooding across all available passes, and $T(t)$ is a spatially uniform scalar estimated from meteorological data available at hourly cadence. The factorisation assigns each term to the data that can actually constrain it. Its cost is an explicit modelling assumption, that the spatial pattern of relative risk does not change with event magnitude, which we return to in Section V-F.

B. SAR flood-frequency inventory

For each monsoon window (1 June–30 September 2024; 15 May–30 September 2025), every Sentinel-1 GRD scene matching the fixed orbit geometry is speckle-filtered with a 30 m focal median and thresholded for open water:

$$W_k(\mathbf{x}) = \mathbb{I}[\sigma_{VV,k}^0(\mathbf{x}) < -16 \text{ dB}] \cdot \mathbb{I}[\sigma_{VH,k}^0(\mathbf{x}) < -22 \text{ dB}], \quad (2)$$

where $\sigma_{VV,k}^0$ and $\sigma_{VH,k}^0$ are calibrated backscatter coefficients for pass k . A pixel counts as *flooded* on pass k only if it is water then and dry in a January–February baseline, and is not permanent water:

$$F_k(\mathbf{x}) = W_k(\mathbf{x}) \cdot (1 - W_{\text{dry}}(\mathbf{x})) \cdot \mathbb{I}[O_{\text{JRC}}(\mathbf{x}) < 80\%], \quad (3)$$

with O_{JRC} the JRC Global Surface Water occurrence [27]. The inventory is the normalised frequency over the $K = 20$ qualifying passes:

$$\Phi(\mathbf{x}) = \frac{1}{K} \sum_{k=1}^K F_k(\mathbf{x}) \in [0, 1]. \quad (4)$$

Vectorising $\Phi > 0$ over the routing window yields 201 polygons totalling 2.86 km² (Figure 3). This area is a *lower bound* on flooding that occurred, not a flood map: it records only what was inundated at the instants the satellite happened to look.

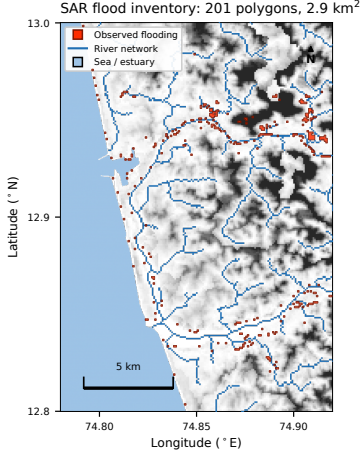


Fig. 3: The multi-temporal SAR flood inventory over the routing window: 201 polygons, 2.86 km², on a HAND backdrop with the channel network and estuary. Observed flooding concentrates in the low-HAND corridors flanking the Gurupura and Nethravathi and along the tidal margin.

C. Conditioning factors and collinearity screening

Twelve candidate factors were computed (Figure 4). TWI and the stream power index follow the standard definitions [21]:

$$\text{TWI} = \ln\left(\frac{a}{\tan \beta}\right), \quad \text{SPI} = \ln(a \tan \beta), \quad (5)$$

with a the specific catchment area (m²) and β the local slope. HAND is the elevation of a pixel above the nearest hydrologically connected drainage cell [22, 23].

Collinearity was screened by variance inflation factor,

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (6)$$

where R_j^2 is the coefficient of determination of factor j regressed on all others. As Table III shows, TWI and SPI are mutually collinear by construction, both being functions of a and $\tan \beta$; with both present they score 16.22 and 14.50. Removing SPI reduces TWI to 1.49 and leaves every retained factor below 3.5, comfortably under the conventional threshold of 10.

D. Sampling with terrain-matched negatives

Training points are drawn by stratified sampling, 1500 per class, under two constraints. Positives require $\Phi(\mathbf{x}) \geq 0.05$; negatives require $\Phi(\mathbf{x}) = 0$ and a distance of at least 120 m from any flood pixel, which removes the ambiguous halo around observed flooding. Both classes are confined to a floodable-lowland domain,

$$\mathcal{D} = \{\mathbf{x} : z(\mathbf{x}) \leq 280 \text{ m} \wedge h(\mathbf{x}) \leq 80 \text{ m}\}, \quad (7)$$

with z elevation and h HAND, both thresholds set at the 90th percentile of the flood-pixel distribution. Without $\mathcal{D}$ the negative class would be drawn largely from forested uplands, and the classifier could separate the classes on elevation alone

without learning anything flood-specific. This elevation- and HAND-matching of negatives is, in our assessment, the single decision that most affects whether the reported scores mean anything.

E. A remote-sensing confound, and its removal

The first model trained on all twelve factors reached a test ROC-AUC of 0.9878. SHAP attribution, however, ranked land cover and NDVI as the two most important predictors, ahead of elevation and HAND (Figure 7c). Inspecting the sample explained why: within the floodable-lowland domain, non-flood pixels are overwhelmingly closed-canopy vegetation. C-band SAR cannot observe standing water beneath a closed canopy [28, 29]. A pixel under forest is therefore labelled “not observed flooded” for a sensing reason, not a hydrological one, and a model given land cover can achieve high accuracy by learning *where SAR can see the ground*, not where water collects.

This is circularity, not signal, and it would have transferred a spurious model into an operational routing system. We therefore define the primary feature set as terrain and rainfall only, nine factors, and retain the twelve-factor model solely as a sensitivity analysis. Section V-A quantifies the cost of this decision: 0.9878 to 0.9625 in test ROC-AUC, with the SHAP ranking reverting to elevation, HAND, distance to river, rainfall and TWI, an ordering consistent with the physical literature.

F. Classifier, calibration and explanation

Random forest [39] and XGBoost [40] were trained on a 70/30 stratified split, with 5-fold cross-validation on the training portion. XGBoost was selected on test ROC-AUC.

Raw tree-ensemble scores are ranking outputs, not probabilities [41], so we calibrate with isotonic regression [42] fitted by cross-validation on the training split alone, and evaluate with the Brier score [43]. Because the training sample is class-balanced while the landscape is not, the calibrated score is tied to a 0.5 prior; Section V-B quantifies this and applies the standard prior correction [44, 45]:

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2, \quad (8)$$

with $\hat{p}_i$ the predicted probability and $y_i \in \{0, 1\}$ the label. Isotonic regression finds the monotone g minimising $\sum_i (g(s_i) - y_i)^2$ over the raw scores s_i , so calibration cannot reorder predictions and therefore cannot alter AUC by construction.

Feature attribution uses SHAP values [10], the Shapley decomposition of a prediction over feature subsets:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|! (|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} [f(S \cup \{j\}) - f(S)], \quad (9)$$

with $\mathcal{F}$ the feature set; we report $|\phi_j|$ over the test split.

TABLE III: Conditioning factors. VIF is reported both with SPI present (11 numeric candidates) and for the retained primary set. SPI is dropped because it and TWI are algebraically coupled through specific catchment area and slope; removing one resolves both.

Factor	Definition / units	VIF (11)	VIF (9)	Source
Elevation	Height above sea level (m)	3.87	3.46	SRTM
Slope	Terrain gradient (ř)	3.72	1.54	SRTM
Aspect	Slope azimuth (ř)	1.04	1.01	SRTM
Curvature	Laplacian of elevation (–)	1.10	1.10	SRTM
HAND	Height above nearest drainage (m)	2.91	2.85	MERIT Hydro
TWI	$\ln(a/\tan\beta)$ (–)	16.22	1.49	MERIT Hydro
Distance to river	Euclidean distance to channel (m)	2.71	2.71	MERIT Hydro
Drainage density	Channel fraction in 1 km neighbourhood (–)	2.21	2.19	MERIT Hydro
Annual rainfall	2020–2024 mean (mm)	2.16	2.14	CHIRPS
<i>Dropped:</i>				
SPI	$\ln(a\tan\beta)$ (–)	14.50	–	VIF > 10
NDVI	Normalised difference vegetation index	1.52	–	Confound (Sec. IV-E)
Land cover	ESA WorldCover class	–	–	Confound (Sec. IV-E)

G. The validation ladder

Three successively stricter protocols are reported.

(i) **Random split.** Standard stratified 70/30 with 5-fold CV. Comparable to published practice, and optimistic for the reasons below.

(ii) **Spatial-block cross-validation.** Samples are assigned to 0.05° (≈ 5.5 km) blocks and whole blocks are held out [15], so no test point shares a neighbourhood with a training point. The block size exceeds the terrain autocorrelation range in this setting.

(iii) **Geographic transfer.** The strictest test. The model is trained *only* on the 2640 labelled points outside the Mangaluru window, predicts wall-to-wall inside it, and is scored against the SAR polygons observed there. The model never sees a label from the evaluation area, which is what deployment to a neighbouring taluk would demand.

For (iii) we report ROC-AUC, hit rate (probability of detection), false-alarm ratio and critical success index:

$$\begin{aligned} \text{POD} &= \frac{\text{TP}}{\text{TP} + \text{FN}}, & \text{FAR} &= \frac{\text{FP}}{\text{TP} + \text{FP}}, \\ \text{CSI} &= \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \end{aligned} \quad (10)$$

together with the success-rate curve [46], the cumulative fraction of observed flooding captured as the landscape is ranked by descending susceptibility. Introduced for landslide-hazard validation and standard in susceptibility mapping since, the success-rate curve is the appropriate headline metric here, because FAR and CSI are severely distorted by the fact that SAR-observed flooding covers only 1.1% of the window and absence of observation is not evidence of dryness.

H. Dynamic trigger

The trigger maps antecedent rainfall to $[0, 1]$ through three terms referenced to India Meteorological Department daily-

rainfall categories, without any parameter fitted to the events it is later tested on:

$$\begin{aligned} T(t) = \min \Big(& \underbrace{0.5 \frac{\min(P_{24}, \theta_{\text{VH}})}{\theta_{\text{VH}}}}_{\text{daily intensity}} + \underbrace{0.3 \frac{\min(P_{72}, \theta_{\text{sat}})}{\theta_{\text{sat}}}}_{\text{catchment saturation}} \\ & + \underbrace{0.2 \frac{\min(P_6, \theta_{\text{H}})}{\theta_{\text{H}}}}_{\text{short burst}}, 1 \Big), \end{aligned} \quad (11)$$

where P_6, P_{24}, P_{72} are trailing rainfall totals (mm) over 6, 24 and 72 h; $\theta_{\text{H}} = 64.5$ mm (IMD Heavy) and $\theta_{\text{VH}} = 115.6$ mm (IMD Very Heavy) are published category thresholds; and $\theta_{\text{sat}} = 200$ mm is a saturation reference. We note explicitly that while θ_{H} and θ_{VH} are published IMD category boundaries, the three weights and θ_{sat} are chosen by us: θ_{sat} is a round value on the order of the wet-season soil-moisture storage of these catchments, not a measured constant. Section V-D sweeps all four. The three-term structure encodes the physical premise that a saturated catchment floods on less additional rain. Equation (11) is monotone non-decreasing in each P by construction, a property we verify numerically rather than assert (Section V-D). The outer $\min(\cdot, 1)$ is redundant at the operating weights, since they sum to one and each term is individually capped, but is retained because Section V-D evaluates weightings for which it binds.

I. Implementation

The pipeline is Python 3.13 with pinned dependencies: `earthengine-api` and `geemap` for server-side processing, `scikit-learn` [47], `xgboost` [40] and `shap` [10] for modelling, `osmnx` [38] and `networkx` for network acquisition and reference validation, `geopandas/rasterio/shapely` for geometry, `numpy` [48] and `scipy` [49] for the statevector simulation and the Hungarian reference solver, and `matplotlib` [50] for all figures. Artifacts are built by a dependency-aware Makefile, so a full rebuild regenerates every number and figure reported here from source data. A suite of 80

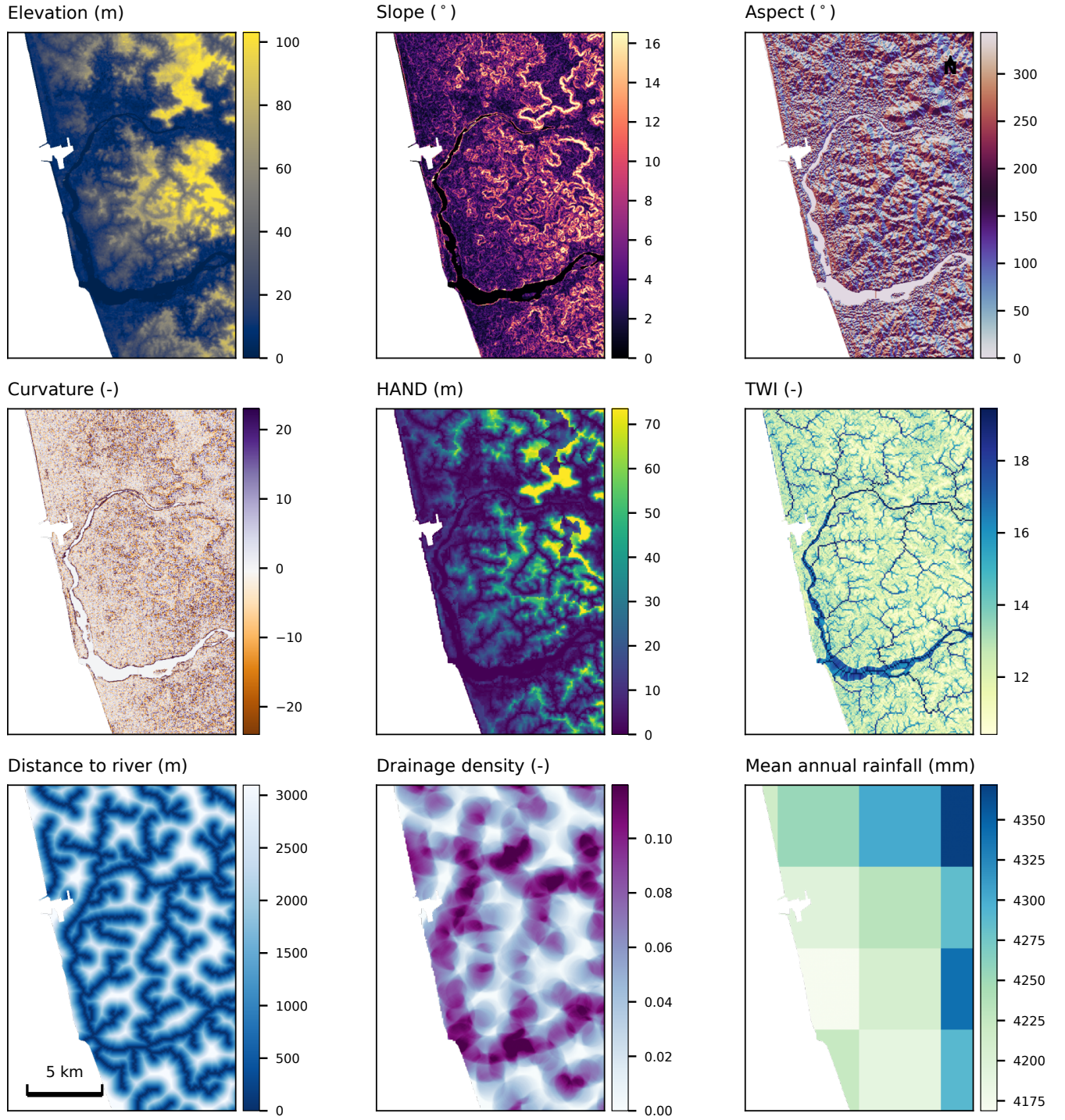


Fig. 4: The nine conditioning factors of the primary model over the Mangaluru window. The channel network is visible as low-slope, high-TWI, low-HAND corridors; the coarse CHIRPS grid is apparent in the rainfall panel and is a known resolution limit of that product.

automated tests encodes the methodological invariants as executable assertions, including class balance and domain confinement of the sample, $VIF < 10$ for all retained factors, monotonicity of $T(t)$, heuristic admissibility, agreement of Dijkstra, A* and networkx on every benchmarked path, and equality of the QUBO ground state with the Hungarian

optimum. Every experiment reported here ran on one Apple M3 Pro workstation with 18GB RAM. Apart from the server-side raster work that Earth Engine performs on its own infrastructure, no additional compute was needed.

V. RESULTS AND DISCUSSION

A. Feature-set ablation: the price of removing circularity

Table IV reports the ablation ladder. The twelve-factor model is the strongest by every metric, and is the model we discard. Its SHAP ranking places land cover first and NDVI second; once both are removed the ranking becomes elevation, HAND, distance to river, annual rainfall, TWI, which is the ordering the flood-susceptibility literature reports on independent study regions [8, 9, 11]. Dropping SPI on collinearity grounds costs essentially nothing (0.9878 to 0.9872), confirming it carried no independent information. Dropping NDVI and land cover costs 0.025 AUC and 0.038 F_1 .

We regard the 0.9625 figure as the honest one and the 0.9878 as an artefact of sensor physics. A reviewer comparing this work against papers reporting 0.98–0.99 on similar data should be aware that the difference may be a methodological choice, not a modelling deficiency.

a) *A smaller version of the same problem, in a factor we retained:* Having built the argument above, it would be inconsistent not to apply the same scepticism to the surviving features, and one does not fully survive it. CHIRPS has a 0.05° (≈ 5.5 km) grid, so the routing window contains only 17 distinct rainfall values across 292,125 pixels, yet SHAP ranks `rain_annual` fourth. A variable that coarse cannot be resolving fine-scale hydrology. Measuring what it is actually doing: its correlation with distance from the coast is 0.73 and with elevation 0.33, so within this window it functions substantially as a smooth coast-to-inland gradient, that is, a proxy for the orographic ramp, not a local rainfall signal.

It is not, however, empty. Leave-one-out retraining shows removing it costs 0.0063 AUC, the second largest of any single factor after elevation (0.0173), so it carries information the terrain variables do not. Our reading is that at district scale the CHIRPS gradient encodes a real orographic precipitation contrast that matters for susceptibility, while at city scale it degenerates into a positional covariate. We retain it, because the model is trained district-wide, and flag it here so the SHAP ranking is not over-interpreted as evidence of fine-scale rainfall control. This is a weaker version of the NDVI problem: a variable earning attribution partly for a reason other than the one its name implies.

Model selection between random forest and XGBoost on the primary set is reported in Table V and Figure 5a; the two are close, and XGBoost is selected on test ROC-AUC. Its SHAP attribution (Figure 5b) ranks elevation and HAND first, followed by channel proximity, rainfall climatology and wetness, an ordering consistent with the hydrological reading of the terms and free of the land-cover dominance that characterised the discarded 12-factor model.

B. Probability calibration, and what the score does not mean

Isotonic calibration improved the Brier score from 0.0697 to 0.0683 and, being a monotone transform, left ranking performance unchanged (ROC-AUC 0.9648). The reliability diagram (Figure 6) tracks the diagonal across the probability range.

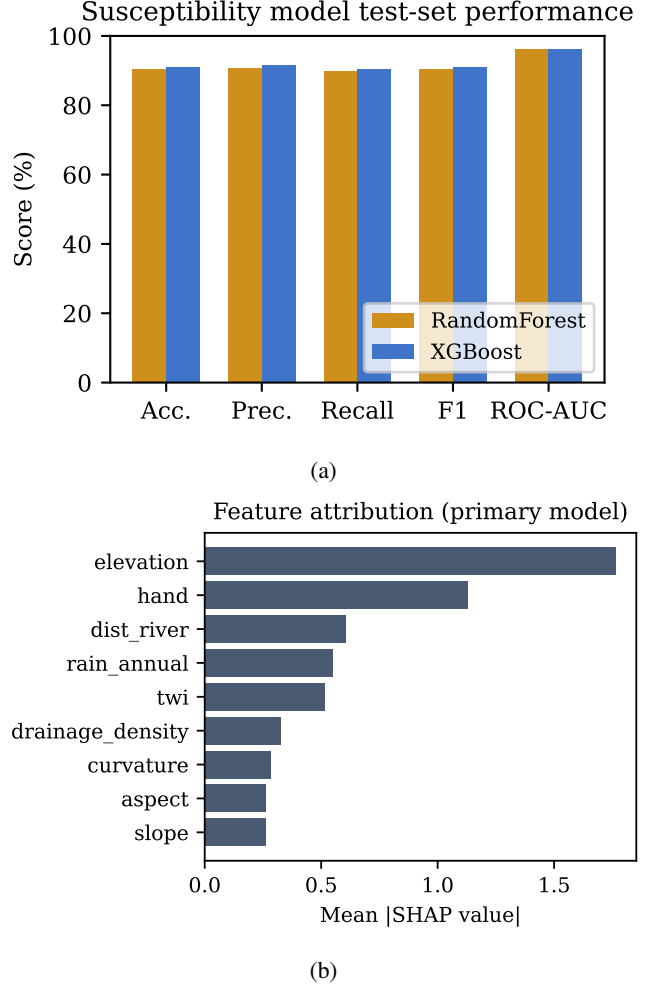


Fig. 5: (a) Random forest versus XGBoost on the primary 9-factor set. (b) SHAP attribution for the primary model: elevation and HAND dominate, followed by channel proximity, rainfall climatology and wetness.

It would be easy, and wrong, to conclude from this that $S(x)$ can be read as a probability of flooding. The training sample is stratified 1500/1500, so its implied base rate is exactly 0.5 by construction, and isotonic regression fitted on that split calibrates to a 0.5 prior. The deployment prevalence, measured as the SAR-observed flooded fraction of the routing window, is 0.0111 (Table VIII). The calibrated score is therefore well-behaved with respect to a reference population in which half the landscape floods, which is not the landscape.

The consequence is measurable and, importantly, it partly explains a result we initially attributed elsewhere. At a 0.5 cut the balanced-prior model flags 34.3% of the window as positive. Applying the standard prior correction [44, 45], for a model trained at prior π_{tr} and deployed at π_{dep} ,

$$p' = \frac{p\omega}{p\omega + (1-p)}, \quad \omega = \frac{\pi_{dep}/(1-\pi_{dep})}{\pi_{tr}/(1-\pi_{tr})} = 0.0112, \quad (12)$$

drops that to 3.66% (Table VI). Because (12) is strictly increasing in p , ROC-AUC is invariant at 0.9648, which is the point:

TABLE IV: Feature-set ablation on the held-out test split ($n = 900$). “12F” includes SPI, NDVI and land cover; “11F” drops SPI on VIF grounds; “9F” is the primary terrain-and-rainfall model. All models are XGBoost with identical hyperparameters and split.

Set	Acc.	Prec.	Rec.	F_1	ROC-AUC	CV AUC	SHAP top-3
12F	0.9456	0.9368	0.9556	0.9461	0.9878	0.9839 ± 0.0092	lulc, ndvi, elevation
11F	0.9478	0.9390	0.9578	0.9483	0.9872	0.9833 ± 0.0098	lulc, ndvi, elevation
9F	0.9111	0.9167	0.9044	0.9105	0.9625	0.9585 ± 0.0154	elevation, hand, dist_river

TABLE V: Primary (9-factor) model performance on the held-out test split ($n = 900$).

Model	Accuracy	Precision	Recall	F_1	ROC-AUC	5-fold CV AUC
Random forest	0.9033	0.9060	0.9000	0.9030	0.9618	0.9554 ± 0.0110
XGBoost	0.9111	0.9167	0.9044	0.9105	0.9625	0.9534 ± 0.0086

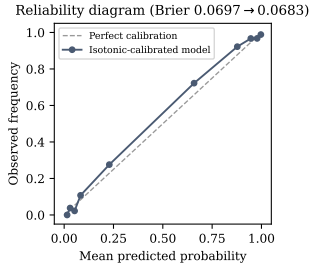


Fig. 6: Reliability diagram after isotonic calibration. Predicted probabilities track observed frequencies across the range.

the ranking was never in question, only the interpretation of its absolute level.

Section V-C defends the high false-alarm ratio on the grounds that absence of SAR observation is not evidence of dryness. That argument stands, but it is not the whole explanation, and we correct it here: a substantial part of the 34.3% positive fraction is simply the balanced-prior artefact. We therefore report $S(x)$ throughout as a *relative exposure ranking*, which is what the routing layer consumes and what the success-rate curve evaluates, and we do not recommend thresholding it at 0.5 as though it were a posterior. An operational deployment wanting absolute probabilities should apply (12) with a locally estimated prevalence, bearing in mind that the SAR-derived π_{dep} used here is itself a lower bound.

C. Spatial validation: how much skill is real

Table VII and Figure 7 report the validation ladder, and the pattern across it is the most methodologically interesting result in this paper.

Under spatial-block cross-validation the primary model falls from 0.9585 to 0.9474, a drop of 1.16%. This is small, and it means the model is not substantially exploiting spatial autocorrelation: held-out geography costs it about one point of AUC. Notably the twelve-factor model drops far *less* (0.30%), which is not evidence that it generalises better. Land cover is so strongly predictive of the SAR label that it swamps the spatial structure, so the block-CV control has little left to remove. A spatial-CV score taken in isolation would therefore

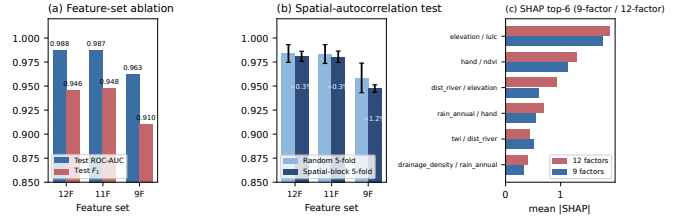


Fig. 7: (a) Feature-set ablation. (b) Random versus spatial-block cross-validation; the confounded 12F model drops least, because land cover dominates the spatial signal. (c) SHAP top-6 for the 9-factor and 12-factor models side by side, showing land cover and NDVI displacing terrain in the confounded model.

have ranked the confounded model as the more robust one, which is a caution worth recording: spatial validation controls for leakage through geography, not for leakage through a sensor artefact.

Under the strictest test, geographic transfer, AUC falls to 0.871 (Table VIII). Hit rate is high at 0.942, and the success-rate curve (Figure 8b) shows the top 10% of the landscape by susceptibility contains 49.5% of all SAR-observed flooding and the top 20% contains 77.6%, lifts of $4.9\times$ and $3.9\times$ over a random ranking. This is the number we would defend operationally. Figure 9 shows the two surfaces side by side: the model trained entirely outside this window reproduces the same drainage-following structure as the one trained on it, which is the qualitative counterpart of that transfer score.

The false-alarm ratio of 0.970 requires care, not concealment. Observed flood pixels are 1.1% of the window, so at a 0.5 threshold the majority of predicted-positive pixels are necessarily unobserved-flooded. More importantly, under a twelve-day revisit a pixel outside the polygons means “not seen flooded”, never “dry”. The FAR is therefore an upper bound on the true false alarm rate and should not be read as a 97% error rate; the AUC and the success-rate curve, both of which use the full ranking rather than a threshold, are the interpretable summaries.

TABLE VI: Effect of the training prior. The classifier is trained on a balanced sample; the landscape is not balanced. The prior-corrected posterior changes what a score means, not how scores rank.

Quantity	Balanced prior	Prevalence-corrected
Implied base rate	0.500	0.0111
Brier score (balanced test split)	0.0683	—
ROC-AUC	0.9648	0.9648 (invariant)
Landscape flagged at a 0.5 cut	34.3%	3.66%
Mean score over the window	0.354	0.071
99th percentile of the window	—	0.678

TABLE VII: Random versus spatial-block 5-fold cross-validation. Blocks are 0.05° squares; whole blocks are held out.

Feature set	Random 5-fold AUC	Spatial-block AUC	Drop	Drop (%)
12F	0.9839 ± 0.0092	0.9810 ± 0.0052	0.0030	0.30
11F	0.9833 ± 0.0098	0.9805 ± 0.0059	0.0028	0.29
9F	0.9585 ± 0.0154	0.9474 ± 0.0039	0.0111	1.16

TABLE VIII: Geographic-transfer validation. The model is trained on the 2640 labelled points *outside* the Mangaluru window and scored against SAR polygons inside it; 360 inside points are withheld entirely. Contingency metrics use a 0.5 threshold.

Quantity	Value	Quantity	Value
Training points (outside)	2640	ROC-AUC	0.871
Points withheld (inside)	360	Hit rate (POD)	0.942
Valid prediction pixels	291,546	False-alarm ratio [†]	0.970
Observed-flood pixels	3227 (1.11%)	Critical success index [†]	0.030
TP / FN	3040 / 187	Flooding in top 10% area	49.5%
TN / FP	191,393 / 96,926	Flooding in top 20% area	77.6%

[†]Upper and lower bounds respectively: SAR observes flooding only on its twelve-day revisit, so a pixel outside the polygons is “not observed flooded”, not “dry”.

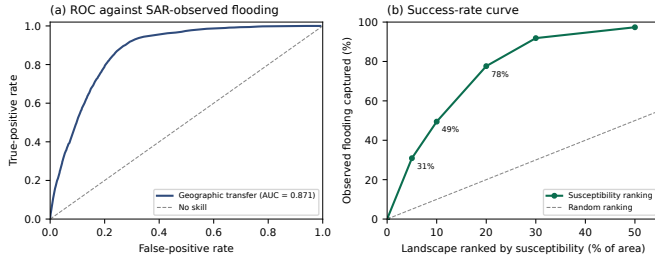


Fig. 8: Geographic transfer. (a) ROC against SAR-observed flooding for a model trained entirely outside the evaluation window. (b) Success-rate curve: the highest-susceptibility 10% of the landscape contains 49.5% of observed flooding.

D. Trigger validation and its discriminative limit

Table IX publishes the complete input record behind every trigger value, which we consider necessary: a trigger whose inputs are not shown cannot be checked. Equation (11) is verified monotone under uniform rainfall scaling ($T = 0.000, 0.063, 0.125, 0.251, 0.502, 0.924, 1.000$ for scale factors 0 to 8).

The ordering test passes. Both documented floods score above an ordinary wet day ($T = 0.133$) and a dry baseline ($T = 0.000$), with no parameter fitted to them. Notably the two events arrive at high T by different routes: May 2025 through

daily intensity (87.2 mm/24 h, IMD Heavy) and August 2024 through saturation (216.9 mm/72 h) despite a merely Moderate 24-hour total. A single-threshold rainfall alarm keyed to 24-hour totals would have ranked August 2024 below an ordinary heavy day; the three-term structure is what recovers it.

The adversarial case is where the result is weaker, and we report it as measured. A deliberately chosen heavy-rain day with no reported flooding (15 July 2025; 66.8 mm/24 h, IMD Heavy) scores $T = 0.551$, only 0.011 below the August 2024 flood. The trigger separates flood from non-flood conditions strongly but separates flood from *heavy-rain-without-flood* barely at all. Figure 10b shows why, and also suggests the remedy: the discriminating variable is the 72-hour saturation total (216.9 and 205.6 mm for the two floods versus 116.8 mm for the non-flood day), while the burst term actively works against separation, being highest of all on the non-flood day (28.0 mm/6 h). Re-weighting toward saturation would widen the margin, but doing so using these events would convert a validated trigger into a fitted one, so we report the limitation rather than tune it away.

Sensitivity of the trigger’s own constants: Equation (11) contains four author-chosen constants: the three weights and θ_{sat} . Only θ_H and θ_{VH} are externally published. Having swept the routing constants, it would be inconsistent not to sweep these, particularly since they govern the paper’s weakest result. Table X reports 231 weight triples on the simplex at 0.05

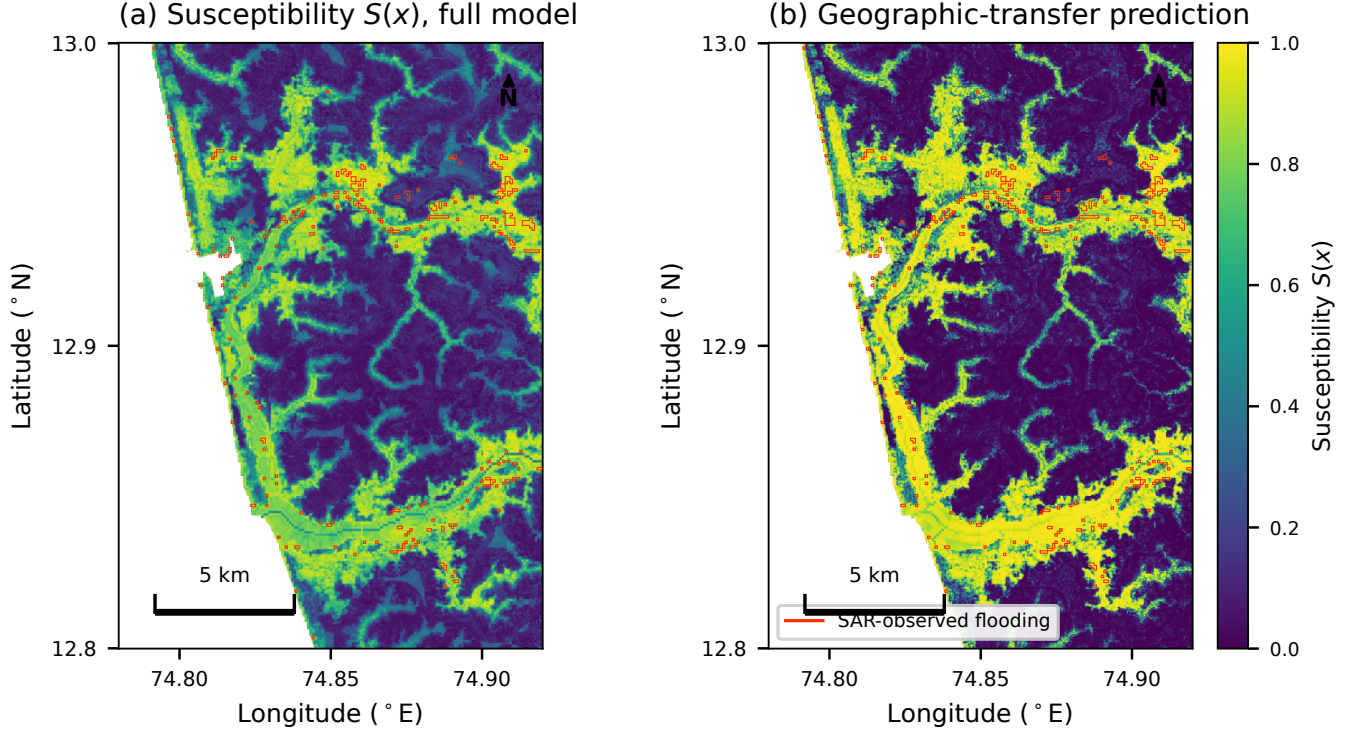


Fig. 9: (a) The calibrated susceptibility surface $S(\mathbf{x})$ over the Mangaluru window with SAR-observed flooding outlined. (b) The geographic-transfer prediction from a model trained outside this window. The two surfaces are visually similar, which is the qualitative counterpart of the transfer AUC of 0.871.

TABLE IX: Complete trigger inputs, from the ERA5 archive via Open-Meteo at Mangaluru (12.87°N, 74.84°E). Weighted terms are the three components of Equation (11). No parameter is fitted to these events.

Scenario	Class	Rainfall (mm)			IMD (24 h)	Weighted terms			T
		6 h	24 h	72 h		daily	sat.	burst	
May 2025 Mangaluru urban	flood	7.6	87.2	205.6	Heavy	0.377	0.300	0.024	0.701
Aug 2024 Nethravathi river	flood	10.1	53.4	216.9	Moderate	0.231	0.300	0.031	0.562
15 Jul 2025 heavy rain	no flood	28.0	66.8	116.8	Heavy	0.289	0.175	0.087	0.551
6 Sep 2025 ordinary wet	wet	3.2	11.0	50.0	Light	0.048	0.075	0.010	0.133
16 Feb 2025 dry baseline	dry	0.0	0.0	0.0	No rain	0.000	0.000	0.000	0.000

resolution crossed with five values of θ_{sat} , evaluated on the same five scenarios.

Two findings. First, the *easy* discrimination is completely robust: floods rank above the ordinary wet day and the dry baseline in all 1155 settings, so that result is a property of the input data, not of the weights. Second, the *hard* discrimination is governed almost entirely by one weight. The adversarial margin rises monotonically from -0.200 at zero saturation weight to $+0.416$ at unit saturation weight (Figure 10c), and is positive in only 374 of 1155 settings. The operating point, at saturation weight 0.3, sits essentially on the zero crossing, which is why its margin is a marginal $+0.011$.

This converts the qualitative diagnosis of the previous subsection into a measurement: the 72-hour antecedent term carries the flood-versus-heavy-rain signal, and the operating weights under-use it. We deliberately do *not* adopt the optimum. Choosing weights to maximise separation on the same

five events the trigger is validated against would convert a validated component into a fitted one and forfeit the property this paper claims for it. Re-weighting requires an independent event set, which we identify as future work.

E. Comparison with prior work

Reported flood-susceptibility AUCs in the literature cluster between 0.85 and 0.95 [6–9, 11]. Our confounded 12-factor model would sit above that range at 0.9878, and our primary model sits inside it at 0.9625. We regard the second number as the comparable one, and note that most reported figures are random-split values whose spatial-block and transfer equivalents are unknown, which is precisely the comparison Section V-C argues the field should be making.

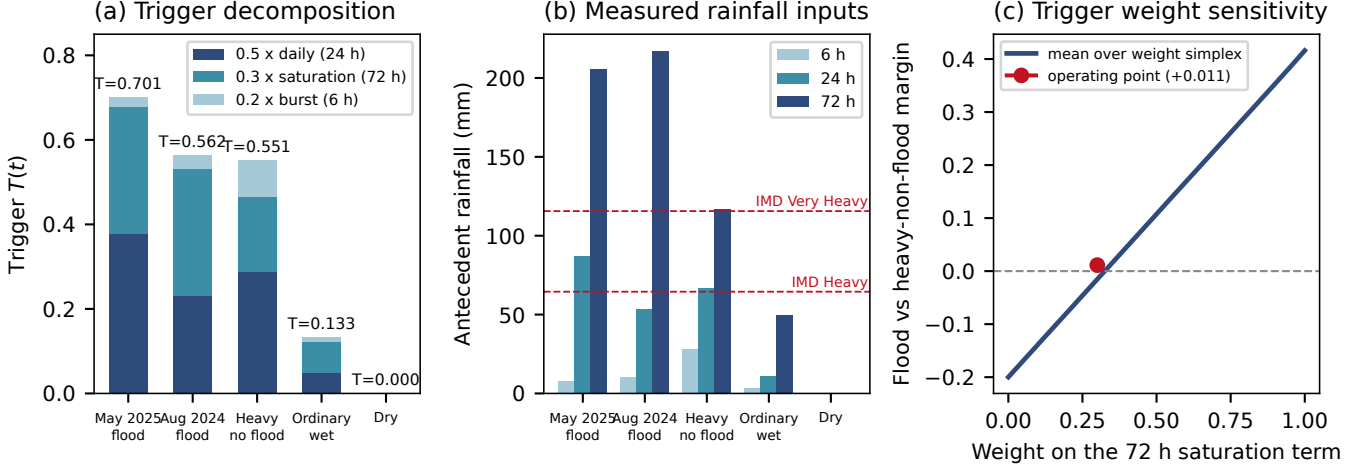


Fig. 10: (a) Decomposition of $T(t)$ into its three weighted terms. (b) The measured rainfall inputs. The two documented floods exceed 200 mm of 72-hour antecedent rainfall; the heavy-rain non-flood day reaches only 116.8 mm despite the largest 6-hour burst in the set, identifying catchment saturation as the discriminating variable. (c) Mean adversarial margin across the weight simplex as a function of the saturation weight, at $\theta_{\text{sat}} = 200$ mm. The operating point sits almost exactly at the zero crossing.

TABLE X: Sensitivity of the trigger to its four unpublished constants, over 231 weight triples $\times$ 5 values of θ_{sat} . “Adversarial margin” is $\min_{\text{floods}} T$ minus T for the heavy-rain non-flood day; “ordinary margin” is against the ordinary wet day. The optimum is reported for information and is not adopted.

Setting	Weights (daily/sat./burst), θ_{sat}	Adversarial	Ordinary
Operating point (adopted)	0.50 / 0.30 / 0.20, 200 mm	+0.011	+0.430
Best over the sweep (not adopted)	0.00 / 1.00 / 0.00, 200 mm	+0.416	+0.610
Worst over the sweep	0.00 / 0.00 / 1.00, 100 mm	−0.316	+0.062
Settings with correct flood > ordinary > dry ordering		1155 / 1155	
Settings with a positive adversarial margin		374 / 1155	
Settings with an adversarial margin above 0.05		277 / 1155	

F. Limitations

- **Susceptibility is not depth.** $S(x)$ and $R(x, t)$ are relative exposure indices in $[0, 1]$, not water depth. A calibrated depth regression, in the direction of Sadhwani and El-dho [31], is the clearest strengthening step.
- **The calibrated score is prior-dependent.** It is calibrated to the balanced training prior, not to landscape prevalence, and must be prior-corrected before being read as a probability.
- **The trigger is a single regional scalar** and uses rainfall only; tide is acquired by the live system but does not enter $T(t)$. Scenarios therefore differ in intensity, not in spatial pattern.
- **The trigger’s free constants are its weakest point.** Three weights and a saturation reference are author-chosen, and the flood versus heavy-rain margin is governed almost entirely by the saturation term.
- **Geographic transfer here is within-district,** across a city boundary rather than across a climatic or geomorphic regime.
- **SAR absence is not evidence of dryness.** The inventory is a lower bound set by a twelve-day revisit.

VI. CONCLUSIONS

This paper built the static half of a two-stage flood-risk model for a district where the satellite record forbids per-event mapping, and then spent most of its effort trying to falsify it.

- 1) **A 12-factor model reaching ROC-AUC 0.9878 was discarded.** SHAP attribution ranked land cover and NDVI first, and the non-flooded lowland of this district is predominantly closed canopy, so the model was learning where SAR can see the ground. Removing the confound costs 0.025 AUC and is mandatory.
- 2) **Stricter validation lowered the number three times.** 0.9625 on a random split, 0.9474 under spatial-block cross-validation, 0.871 on geographic transfer. The last of these is the one an operational deployment would experience, and its highest-susceptibility tenth still contains 49.5% of observed flooding.
- 3) **The calibrated score is not a flood probability.** Isotonic calibration on a balanced sample targets a 50% prior against a landscape prevalence of 1.11%; the correction changes what the number means without changing the ranking.
- 4) **The trigger orders real events without being fitted to them,** but separates a documented flood from a heavy-

rain non-flood day by only 0.011, and the diagnostic identifies the 72-hour saturation term as the variable carrying the signal.

The broader claim is methodological: each protocol we applied removed a specific, nameable way the previous figure was optimistic, and the sequence is more useful to a practitioner than its first term. Part II [18] takes the resulting risk surface into the emergency-response layer, where the same discipline is applied to routing and dispatch.

REFERENCES

- [1] V. D. Chandrashekar, A. Shetty. Trends in extreme rainfall over ecologically sensitive Western Ghats and coastal regions of Karnataka: an observational assessment. *Arabian Journal of Geosciences*, 11:327, 2018. doi:10.1007/s12517-018-3700-6.
- [2] S. Chandana, M. C. Chandan. Hydrological implications of LULC changes in the Netravathi river basin: insights from SWAT modeling. In *2023 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, pages 1–4, 2023. doi:10.1109/InGARSS59135.2023.10490324.
- [3] S. Babar, H. Ramesh. Streamflow response to land use–land cover change over the Nethravathi River basin, India. *Journal of Hydrologic Engineering*, 20(10):05015002, 2015. doi:10.1061/(ASCE)HE.1943-5584.0001177.
- [4] N. Naik, K. Chandrasekaran, V. M. Sundaram, P. Panneer. Dual attention guided deep encoder-decoder network for change analysis in land use/land cover for Dakshina Kannada District, Karnataka, India. *Environmental Earth Sciences*, 82(1):33, 2022. doi:10.1007/s12665-022-10713-1.
- [5] R. Nirmala. Flood susceptibility assessment of Nethravathi River basin, Karnataka, through GIS-AHP technique. In *Sustainability Solutions*, pages 643–661. Springer, 2026. doi:10.1007/978-3-032-11300-9_27.
- [6] M. S. Tehrani, B. Pradhan, M. N. Jebur. Flood susceptibility mapping using a novel ensemble weights-of-evidence and support vector machine models in GIS. *Journal of Hydrology*, 512:332–343, 2014.
- [7] K. Khosravi, B. T. Pham, K. Chapi, A. Shirzadi, H. Shahabi, I. Revhaug, I. Prakash, D. T. Bui. A comparative assessment of decision trees algorithms for flash flood susceptibility modeling at Haraz watershed, northern Iran. *Science of the Total Environment*, 627:744–755, 2018.
- [8] A. Arabameri, K. Rezaei, A. Cerdà, C. Conoscenti, Z. Kalantari. A comparison of statistical methods and multi-criteria decision making to map flood hazard susceptibility in Northern Iran. *Science of the Total Environment*, 660:443–458, 2019.
- [9] A. Saha, S. Pal, A. Arabameri, T. Blaschke, S. Panahi, I. Chowdhuri, R. Chakraborty, R. Costache, A. Arora. Flood susceptibility assessment using novel ensemble of hyperpipes and support vector regression algorithms. *Water*, 13(2):241, 2021. doi:10.3390/w13020241.
- [10] S. M. Lundberg, S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, 2017.
- [11] H. E. Aydin, M. C. Iban. Predicting and analyzing flood susceptibility using boosting-based ensemble machine learning algorithms with SHapley Additive exPlanations. *Natural Hazards*, 116:2957–2991, 2023. doi:10.1007/s11069-022-05793-y.
- [12] S. T. Seydi, Y. Kanani-Sadat, M. Hasanlou, R. Sahraei, J. Chanussot, M. Amani. Comparison of machine learning algorithms for flood susceptibility mapping. *Remote Sensing*, 15(1):192, 2023.
- [13] R. Torres, P. Snoeij, D. Geudtner, et al. GMES Sentinel-1 mission. *Remote Sensing of Environment*, 120:9–24, 2012. doi:10.1016/j.rse.2011.05.028.
- [14] M. Arora, S. Sahoo, C. M. Bhatt, P. K. Litoria, B. Pateriya. Rapid flood inundation mapping and impact assessment using Sentinel-1 SAR data over Ghaggar River basin of Punjab, India. *Journal of Earth System Science*, 132(4):183, 2023. doi:10.1007/s12040-023-02199-7.
- [15] D. R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.
- [16] P. Ploton et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11:4540, 2020.
- [17] H. Meyer, E. Pebesma. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12(9):1620–1633, 2021.
- [18] Authors. Spatiotemporal flood risk modelling under a single-orbit SAR constraint, Part II: Flood-aware emergency routing and hybrid quantum dispatch. *Companion paper, submitted*.
- [19] Y. Wang, Z. Fang, H. Hong, L. Peng. Flood susceptibility mapping using convolutional neural network frameworks. *Journal of Hydrology*, 582:124482, 2020.
- [20] M. Avand, H. Moradi, M. Ramazanzadeh Lasboyee. Using machine learning models, remote sensing, and GIS to investigate the effects of changing climates and land uses on flood probability. *Journal of Hydrology*, 595:125663, 2021.
- [21] K. J. Beven, M. J. Kirkby. A physically based, variable contributing area model of basin hydrology. *Hydrological Sciences Bulletin*, 24(1):43–69, 1979.
- [22] C. D. Rennó, A. D. Nobre, L. A. Cuartas, J. V. Soares, M. G. Hodnett, J. Tomasella, M. J. Waterloo. HAND, a new terrain descriptor using SRTM-DEM: mapping terra-firme rainforest environments in Amazonia. *Remote Sensing of Environment*, 112(9):3469–3481, 2008.
- [23] A. D. Nobre, L. A. Cuartas, M. Hodnett, C. D. Rennó, G. Rodrigues, A. Silveira, M. Waterloo, S. Saleska. Height Above the Nearest Drainage: a hydrologically relevant new terrain model. *Journal of Hydrology*, 404(1–2):13–29, 2011.
- [24] S. Martinis, A. Twele, S. Voigt. Towards operational near real-time flood detection using a split-based automatic thresholding procedure on high resolution TerraSAR-X

- data. *Natural Hazards and Earth System Sciences*, 9:303–314, 2009.
- [25] A. Tuele, W. Cao, S. Plank, S. Martinis. Sentinel-1-based flood mapping: a fully automated processing chain. *International Journal of Remote Sensing*, 37(13):2990–3004, 2016.
- [26] X. Shen, D. Wang, K. Mao, E. Anagnostou, Y. Hong. Inundation extent mapping by synthetic aperture radar: a review. *Remote Sensing*, 11(7):879, 2019.
- [27] J.-F. Pekel, A. Cottam, N. Gorelick, A. S. Belward. High-resolution mapping of global surface water and its long-term changes. *Nature*, 540:418–422, 2016.
- [28] P. A. Townsend. Relationships between forest structure and the detection of flood inundation in forested wetlands using C-band SAR. *International Journal of Remote Sensing*, 23(3):443–460, 2002.
- [29] V. Tsyganskaya, S. Martinis, P. Marzahn, R. Ludwig. SAR-based detection of flooded vegetation: a review of characteristics and approaches. *International Journal of Remote Sensing*, 39(8):2255–2293, 2018.
- [30] G. Donchyts, F. Baart, H. Winsemius, N. Gorelick, J. Kwadijk, N. van de Giesen. Earth’s surface water change over the past 30 years. *Nature Climate Change*, 6:810–813, 2016.
- [31] K. Sadhwani, T. I. Eldho. Machine learning-driven flood hazard assessment: integrating SAR and elevation data for inundation mapping and depth estimation. *Natural Hazards*, 122:177, 2026. doi:10.1007/s11069-025-07949-y.
- [32] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, R. Moore. Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202:18–27, 2017.
- [33] T. G. Farr et al. The Shuttle Radar Topography Mission. *Reviews of Geophysics*, 45(2):RG2004, 2007.
- [34] D. Yamazaki, D. Ikeshima, J. Sosa, P. D. Bates, G. H. Allen, T. M. Pavelsky. MERIT Hydro: a high-resolution global hydrography map based on latest topography datasets. *Water Resources Research*, 55(6):5053–5073, 2019.
- [35] C. Funk et al. The climate hazards infrared precipitation with stations: a new environmental record for monitoring extremes. *Scientific Data*, 2:150066, 2015.
- [36] D. Zanaga et al. ESA WorldCover 10m 2020 v100. Zenodo, 2021. doi:10.5281/zenodo.5571936.
- [37] H. Hersbach, B. Bell, P. Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi:10.1002/qj.3803.
- [38] G. Boeing. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65:126–139, 2017. doi:10.1016/j.compenvurbsys.2017.05.004.
- [39] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- [40] T. Chen, C. Guestrin. XGBoost: A scalable tree boosting system. In *Proc. 22nd ACM SIGKDD*, pages 785–794, 2016.
- [41] A. Niculescu-Mizil, R. Caruana. Predicting good probabilities with supervised learning. In *Proc. 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005. doi:10.1145/1102351.1102430.
- [42] B. Zadrozny, C. Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proc. 8th ACM SIGKDD*, pages 694–699, 2002.
- [43] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- [44] C. Elkan. The foundations of cost-sensitive learning. In *Proc. 17th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 973–978, 2001.
- [45] M. Saerens, P. Latinne, C. Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002. doi:10.1162/089976602753284446.
- [46] C.-J. F. Chung, A. G. Fabbri. Validation of spatial prediction models for landslide hazard mapping. *Natural Hazards*, 30(3):451–472, 2003. doi:10.1023/B:NHAZ.0000007172.62651.2b.
- [47] F. Pedregosa, G. Varoquaux, A. Gramfort, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [48] C. R. Harris, K. J. Millman, S. J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi:10.1038/s41586-020-2649-2.
- [49] P. Virtanen, R. Gommers, T. E. Oliphant, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi:10.1038/s41592-019-0686-2.
- [50] J. D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi:10.1109/MCSE.2007.55.